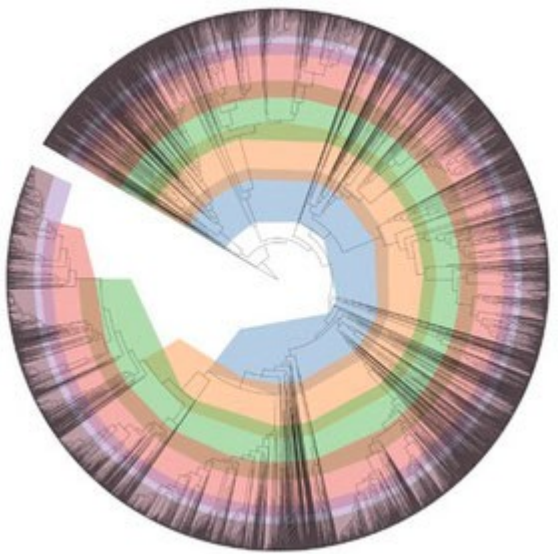
ENVIRONMENTAL METAGENOMICS

Physalia course

Course outline and practical information

Nikolay Oskolkov, Group Leader of Metabolic Research Group at NIRI, Riga, Latvia
Luis Pedro Coelho, Group Leader at Queensland University of Technology, Brisbane, Australia



Physalia Courses

NB: original course material courtesy:
Dr. Antti Karkman, University of Helsinki
Dr. Igor Pessi, Finnish Environment Institute (SYKE)

About us

Organizer: Carlo Pecoraro, Physalia courses

info@physalia-courses.org



Instructors:



Dr. Nikolay Oskolkov, Group Leader at NIRI, Riga, Latvia

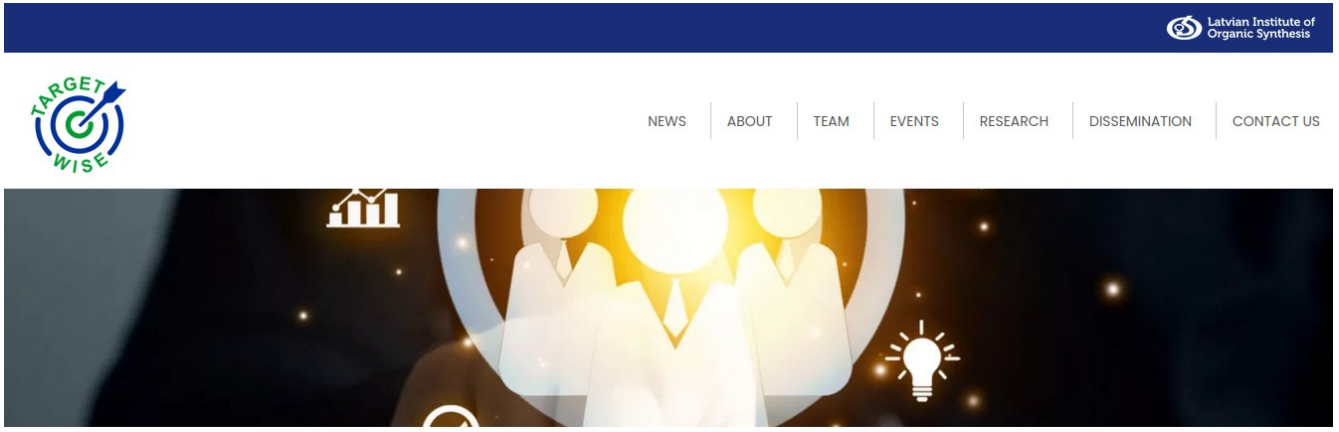
nikolay.oskolkov@niri.lv



Dr. Luis Pedro Coelho, Group leader at the Queensland University of Technology, Brisbane, Australia

luispedro.coelho@qut.edu.au

Metabolic Research Group at NIRI



1 more postdoctoral fellow and 1 PhD student to be hired

If you know anyone who might be interested, please contact me!



PhD. Nikolay Oskolkov
Group Leader (PI) of the Metabolic Research Group



Mohammad Banabazi, postdoc in Metabolic Research Group



Kristina Grausa, PhD student in Metabolic Research Group



Daniel Rivas, MD, postdoc in Metabolic Research Group

Brief introduction: who am I

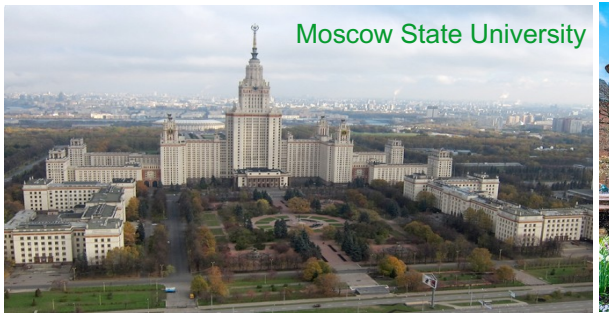
2007 PhD in theoretical physics in Moscow, Russia

2011 medical genomics at Lund University, Sweden

2016 bioinformatician at NBIS SciLifeLab, Sweden

2025 Metabolic Research Group leader, NIRI, Latvia

60+ publications; h-index = 28; 4,600+ citations

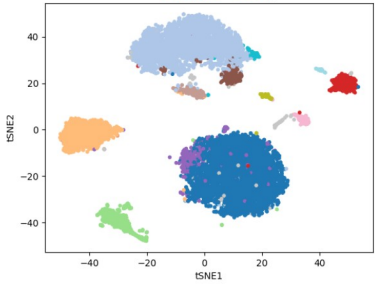








Single cell



Biomedical data integration

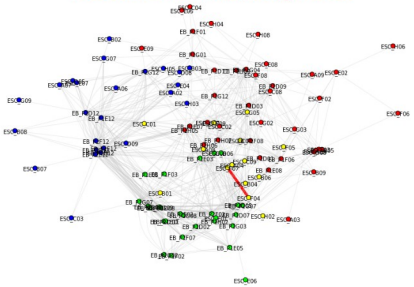
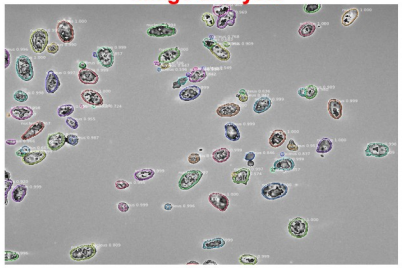
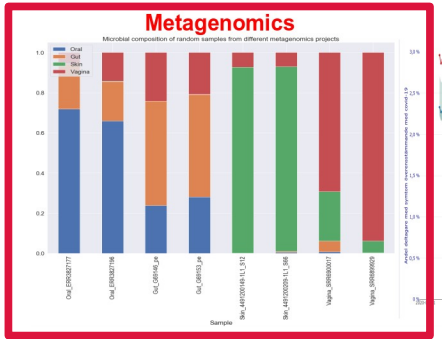


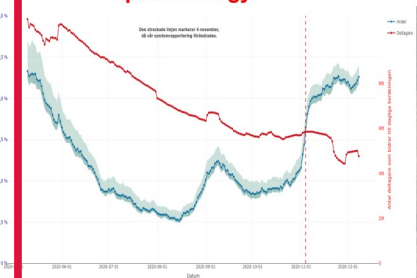
Image analysis



Metagenomics



Epidemiology



Ancient microbiome



aMeta: Pochon et al. Genome Biology 2023

Pochon *et al. Genome Biology* (2023) 24:242
https://doi.org/10.1186/s13059-023-03083-9

Genome Biology

METHOD Open Access

aMeta: an accurate and memory-efficient ancient metagenomic profiling workflow

Zoé Pochon[1,2†], Nora Bergfeldt[1,3,4†], Emrah Kırdök[5], Mário Vicente[1,2], Thijessen Naidoo[1,2,6,7], Tom van der Valk[1,4], N. Ezgi Altınışık[8], Maja Krzewińska[1,2], Love Dalén[1,3], Anders Götherström[1,2†], Claudio Mirabello[9†], Per Unneberg[10†] and Nikolay Oskolkov[11*†]

†Zoé Pochon, Nora Bergfeldt, Anders Götherström, Claudio Mirabello, Per Unneberg, and Nikolay Oskolkov shared authorship.

*Correspondence: Nikolay.Oskolkov@biol.lu.se

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Full list of author information is available at the end of the article

Abstract

Analysis of microbial data from archaeological samples is a growing field with great potential for understanding ancient environments, lifestyles, and diseases. However, high error rates have been a challenge in ancient metagenomics, and the availability of computational frameworks that meet the demands of the field is limited. Here, we propose aMeta, an accurate metagenomic profiling workflow for ancient DNA designed to minimize the amount of false discoveries and computer memory requirements. Using simulated data, we benchmark aMeta against a current state-of-the-art workflow and demonstrate its superiority in microbial detection and authentication, as well as substantially lower usage of computer memory.

Keywords: Ancient metagenomics, Pathogen detection, Microbiome profiling, Ancient DNA

Background

Historically, ancient DNA (aDNA) studies have focused on human and faunal evolution and demography, extracting and analyzing predominantly eukaryotic aDNA [1–3]. With the development of next-generation sequencing (NGS) technologies, it was demonstrated that host-associated microbial aDNA from eukaryotic remains, which was previously treated as a sequencing by-product, can provide valuable information about ancient pandemics, lifestyle, and population migrations in the past [4–6]. Modern technologies have made it possible to study not only ancient microbiomes populating eukaryotic hosts, but also sedimentary ancient DNA (sedaDNA), which has rapidly become an independent branch of palaeogenetics, delivering unprecedented information about hominin and animal evolution without the need to analyze historical bones and teeth [7–12]. Previously available in microbial ecology, meta-barcoding methods lack validation and authentication power, and therefore, shotgun metagenomics has become the *de facto* standard in ancient microbiome research [13]. However, accurate detection,

README MIT license

aMeta

ancient Metagenomic profiling workflow

aMeta: an accurate and memory-efficient ancient Metagenomic profiling workflow

snakemake ≥6.10.0 Tests passing

About

aMeta is a Snakemake workflow for identifying microbial sequences in ancient DNA shotgun metagenomics samples. The workflow performs:

- trimming adapter sequences and removing reads shorter than 30 bp with Cutadapt
- quaity control before and after trimming with FastQC and MultiQC
- taxonomic sequence kmer-based classification with KrakenUniq
- sequence alignment with Bowtie2 and screening for common microbial pathogens
- deamination pattern analysis with MapDamage2
- Lowest Common Ancestor (LCA) sequence alignment with Malt
- authentication and validation of identified microbial species with MaltExtract

When using aMeta and / or pre-built databases provided together with the wokflow for your research projects, please cite our preprint: https://www.biorxiv.org/content/10.1101/2022.10.03.510579v1

Authors

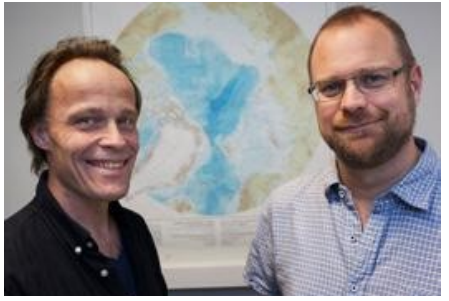
- Nikolay Oskolkov (@LeandroRitter) nikolay.oskolkov@scilifelab.se
- Claudio Mirabello (@clami66) claudio.mirabello@scilifelab.se
- Per Unneberg (@percyfal) per.unneberg@scilifelab.se

https://github.com/NBISweden/aMeta









GENEX: Oskolkov et al. GigaScience 2025

GigaScience, 2025, 14, 1–14
DOI: 10.1093/gigascience/giaf108
Technical Note

Improving taxonomic inference from ancient environmental metagenomes by masking microbial-like regions in reference genomes

Nikolay Oskolkov[1,*], Chenyu Jin[2,3,4], Samantha López Clinton[2,3,4], Benjamin Guinet[2,3], Flore Wijnands[2,5], Ernst Johnson[2,5], Verena E. Kutschera[6], Cormac M. Kinsella[3,7], Peter D. Heintzman[2,5], and Tom van der Valk[2,3,8]

[1]Department of Biology, National Bioinformatics Infrastructure Sweden, Science for Life Laboratory, Lund University, SE-223 62 Lund, Sweden
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[5]Department of Geological Sciences, Stockholm University, SE-106 91 Stockholm, Sweden
[6]Department of Biochemistry and Biophysics, National Bioinformatics Infrastructure Sweden, Science for Life Laboratory, Stockholm University, Solna, SE-106 91 Stockholm, Sweden
[7]Department of Cell and Molecular Biology, National Bioinformatics Infrastructure Sweden, Science for Life Laboratory, Uppsala University, SE-751 24 Uppsala, Sweden
[8]Scilifelab, Tomtebodavägen 23, SE-171 65 Solna, Stockholm, Sweden
*Correspondence address. Nikolay Oskolkov, Department of Biology, National Bioinformatics Infrastructure Sweden, Science for Life Laboratory, Lund University, SE-223 62 Lund, Sweden. E-mail: Nikolay.Oskolkov@biol.lu.se

Abstract

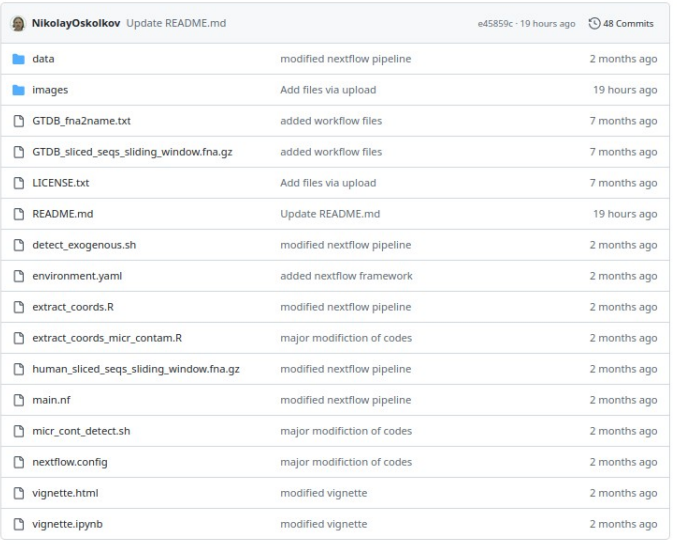
Ancient environmental DNA is increasingly vital for reconstructing past ecosystems, particularly when paleontological and archaeological tissue remains are absent. Detecting ancient plant and animal DNA in environmental samples relies on using extensive eukaryotic reference genome databases for profiling metagenomics data. However, many eukaryotic genomes contain regions with high sequence similarity to microbial DNA, which can lead to the misclassification of bacterial and archaeal reads as eukaryotic. This issue is especially problematic in ancient eDNA datasets, where plant and animal DNA is typically present at very low abundance. In this study, we present a method for identifying bacterial- and archaeal-like sequences in eukaryotic genomes and apply it to nearly 3,000 reference genomes from NCBI RefSeq and GenBank (vertebrates, invertebrates, plants) as well as the 1,323 PhyloNorway plant genome assemblies from herbarium material from northern high-latitude regions. We find that microbial-like regions are widespread across eukaryotic genomes and provide a comprehensive resource of their genomic coordinates and taxonomic annotations. This resource enables the masking of microbial-like regions during profiling analyses, thereby improving the reliability of ancient environmental metagenomic datasets for downstream analyses.

Keywords: environmental DNA, ancient metagenomics, microbial-like regions

Introduction

Ancient environmental DNA (aeDNA) is a tool for studying past ecosystems, especially in contexts where traditional archaeological and paleontological tissue remains, such as bones and seeds, are absent [1–4]. It consists of genetic traces left by organisms in the environment, such as soil, sediments, or ice, and allows for the reconstruction of past biodiversity and ecological communities to provide insight into species extinction, vegetation changes, and ecosystem responses to climatic shifts and anthropogenic impacts.

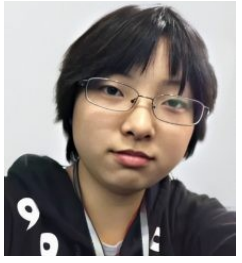
to genomic reference databases. Consequently, the quality of both the aeDNA data and the reference databases is crucial for reliable inferences. Microbial-like sequences in reference genomic databases, originating either from nonendogenous sources (contamination) or from evolutionary similarity to microbial genomes (e.g., due to ancient horizontal gene transfer or the endosymbiotic origins of plastids), can be a potential source of false-positive taxonomic identifications. In such cases, microbial sequences present in aeDNA data may be mistakenly classified as belonging to a eukaryotic reference genome due to sequence similarity.



https://github.com/NikolayOskolkov/MCWorkflow









Published in



Frontiers in Microbiology
Ancient DNA and Forensic Microbiology

5.8 impact factor **9.9** citescore

Part of a Research Topic

Ancient DNA and the History of Human Populations

38k views 4 articles

> Editor & Reviewers

ORIGINAL RESEARCH article

Front. Microbiol., 18 May 2026

Sec. Ancient DNA and Forensic Microbiology

Volume 17 - 2026 | https://doi.org/10.3389/fmicb.2026.1603339

Refining filtering criteria of Kraken family of tools for accurate taxonomic profiling of ancient metagenomic data

Nikolay Oskolkov [1,2]*

1. Department of Biology, National Bioinformatics Infrastructure Sweden, Science for Life Laboratory, Lund University, Lund, Sweden
2. Metabolic Research Group, Latvian Institute of Organic Synthesis, Riga, Latvia

Article metrics **2** Citations **2,1k** Views **643** Downloads View details

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About you

- Name
- University/Institute/Company
- Research interest(s)
- Previous experience(s) with microbial ecology, metagenomics, bioinformatics, etc.
- General hopes for this course

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Course outline

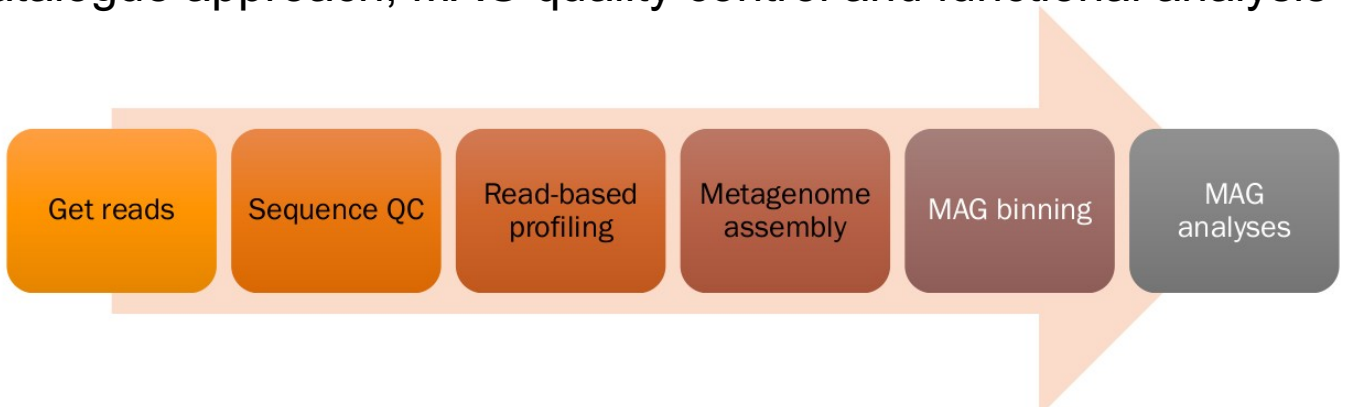
Day 1: introduction, setting up, connecting to server, getting raw data, quality control

Day2: adapter and host removal, decontamination, read-based taxonomic profiling

Day3: *de-novo* assembly, taxonomic profiling and abundance quantification of contigs

Day4: assembly of long-read sequencing data, metagenomic binning

Day5: gene catalogue approach, MAG quality control and functional analysis of MAGs



Physalia Courses

PROGRAM

Monday (9 am - 1 pm Berlin time)

- Course introduction and practical info
- Introduction to metagenomics
- Setting up cloud computing (hands-on)
- Retrieving raw data (hands-on)

Tuesday (9 am - 1 pm Berlin time)

- Q&A from previous day
- Quality control and trimming (slides & hands-on)
- Metagenome assembly (slides & hands-on)
- Assembly quality control (hands-on)
- Read-based taxonomic profiling (slides & hands-on)

Wednesday (9 am - 1 pm Berlin time)

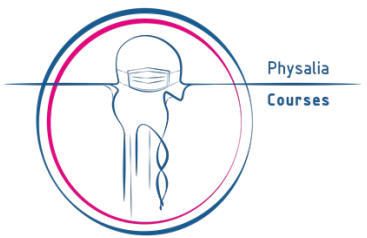
- Q&A from previous day
- Continued read-based taxonomic profiling (slides & hands-on)
- Genome-resolved metagenomics using anvi'o (slides & hands-on)

Thursday (9 am - 1 pm Berlin time)

- Q&A from previous day
- Continued genome-resolved metagenomics using anvi'o (slides & hands-on)
- Continued read-based taxonomic profiling (slides & hands-on)

Friday (9 am - 1 pm Berlin time)

- Continued read-based taxonomic profiling (slides & hands-on)
- Quality control and taxonomic annotation of MAGs (hands-on)
- Automatic binning with SemiBin2 (hands-on)
- Sharing MAG information
- Targeted functional analysis of MAGs (hands-on)
- Open Q&A and discussion



Housekeeping rules

- To ask question please raise your hand, unmute yourself and ask. You can also ask questions in zoom chat or slack workspace for this course.

- Please keep your camera on as much as possible for better contact and communication.

- The course includes ~10 lectures (30–60 min each) followed by practical sessions, we will have 10-15 minutes breaks between the sessions depending on how tired the participants are.

- Recordings will be provided after each day, so if you miss a lecture or practical, there will be a chance to catch up

Physalia
Courses

Practical information: GitHub and Zoom

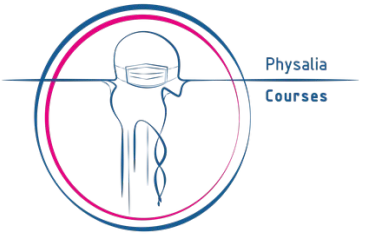
The course will take place in Zoom from 9 am to 1 pm CET, Berlin time

Links to the Zoom room will be posted in Slack

The course GitHub repository containing lectures and exercises is:

https://github.com/NikolayOskolkov/EnvMetagenomics

Please bookmark this address!



Practical information: Amazon Cloud (AWS EC2)

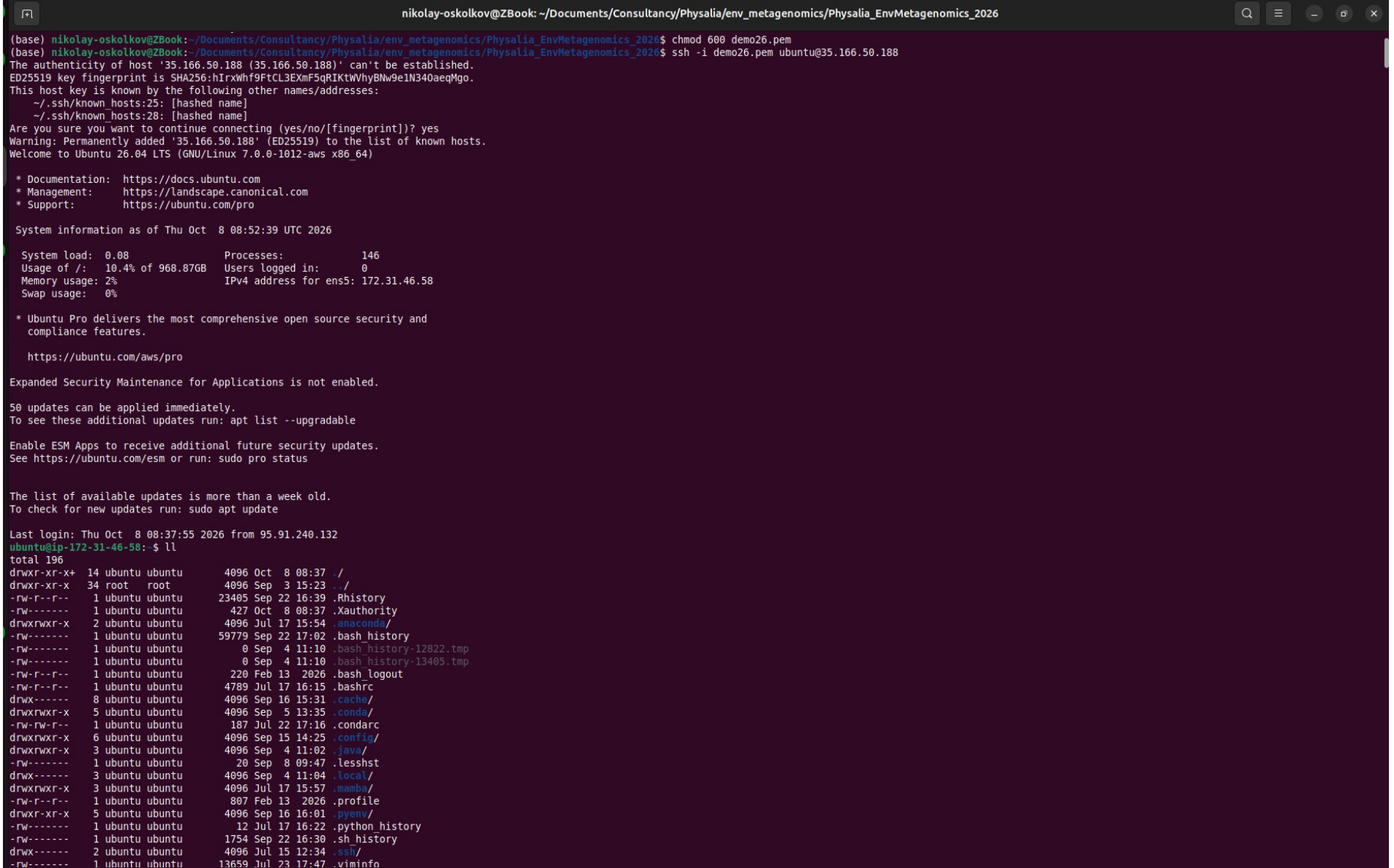
We will use the Cloud Computing service from Amazon, which we will access via `ssh` (secure shell protocol)

https://github.com/NikolayOskolkov/EnvMetagenomics/blob/main/exercises.md#setting-up-the-cloud-computing

See here for information on how to connect, but remember:

- The IP address changes every day
- Everyone is given a username, with a `home` and `shared` folders
  - List of usernames can be found in Slack
  - The `shared` folder is copy-only: do not delete, move, rename, or write

Physalia Courses


nikolay-oskolkov@ZBook: ~/Documents/Consultancy/Physalia/env_metagenomics/Physalia_EnvMetagenomics_2026
(base) nikolay-oskolkov@ZBook:~/Documents/Consultancy/Physalia/env_metagenomics/Physalia_EnvMetagenomics_2026$ chmod 600 demo26.pem
(base) nikolay-oskolkov@ZBook:~/Documents/Consultancy/Physalia/env_metagenomics/Physalia_EnvMetagenomics_2026$ ssh -i demo26.pem ubuntu@35.166.50.188
The authenticity of host '35.166.50.188 (35.166.50.188)' can't be established.
ED25519 key fingerprint is SHA256:hIrxWhf9FtCL3EXmF5qRIKtWVhyBNw9e1N34OaeqMgo.
This host key is known by the following other names/addresses:
~/.ssh/known_hosts:25: [hashed name]
~/.ssh/known_hosts:28: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '35.166.50.188' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1012-aws x86_64)
* Documentation: https://docs.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
System information as of Thu Oct 8 08:52:39 UTC 2026
System load: 0.08
Usage of /: 10.4% of 968.87GB
Memory usage: 2%
Swap usage: 0%
Processes: 146
Users logged in: 0
IPv4 address for ens5: 172.31.46.58
* Ubuntu Pro delivers the most comprehensive open source security and compliance features.
https://ubuntu.com/aws/pro
Expanded Security Maintenance for Applications is not enabled.
50 updates can be applied immediately.
To see these additional updates run: apt list --upgradable
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Last login: Thu Oct 8 08:37:55 2026 from 95.91.240.132
ubuntu@ip-172-31-46-58:~$ ll
total 196
drwxr-xr-x+ 14 ubuntu ubuntu 4096 Oct 8 08:37 ./
drwxr-xr-x 34 root root 4096 Sep 3 15:23 ../
-rw-r--r-- 1 ubuntu ubuntu 23405 Sep 22 16:39 .Rhistory
-rw------- 1 ubuntu ubuntu 427 Oct 8 08:37 .Xauthority
drwxrwxr-x 2 ubuntu ubuntu 4096 Jul 17 15:54 .anaconda/
-rw------- 1 ubuntu ubuntu 59779 Sep 22 17:02 .bash_history
-rw------- 1 ubuntu ubuntu 0 Sep 4 11:10 .bash_history-12822.tmp
-rw------- 1 ubuntu ubuntu 0 Sep 4 11:10 .bash_history-13405.tmp
-rw-r--r-- 1 ubuntu ubuntu 220 Feb 13 2026 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 4789 Jul 17 16:15 .bashrc
drwx------ 8 ubuntu ubuntu 4096 Sep 16 15:31 .cache/
drwxrwxr-x 5 ubuntu ubuntu 4096 Sep 5 13:35 .conda/
-rw-rw-r-- 1 ubuntu ubuntu 187 Jul 22 17:16 .condarc
drwxrwxr-x 6 ubuntu ubuntu 4096 Sep 15 14:25 .config/
drwxrwxr-x 3 ubuntu ubuntu 4096 Sep 4 11:02 .java/
-rw------- 1 ubuntu ubuntu 20 Sep 8 09:47 .lesshst
drwx------ 3 ubuntu ubuntu 4096 Sep 4 11:04 .local/
drwxrwxr-x 3 ubuntu ubuntu 4096 Jul 17 15:57 .mamba/
-rw-r--r-- 1 ubuntu ubuntu 807 Feb 13 2026 .profile
drwxr-xr-x 5 ubuntu ubuntu 4096 Sep 16 16:01 .pyenv/
-rw------- 1 ubuntu ubuntu 12 Jul 17 16:22 .python_history
-rw------- 1 ubuntu ubuntu 1754 Sep 22 16:30 .sh_history
drwx------ 2 ubuntu ubuntu 4096 Jul 15 12:34 .ssh/
-rw------- 1 ubuntu ubuntu 13659 Jul 23 17:47 .viminfo

Practical information: conda

conda.io

System for software management (python, R, JavaScript, C++, ...)

Allows easy installation of software in dedicated environments, separated from the main environment and other conda environments

- The environments that we will use have been already set up for everyone

General conda commands

```
> conda env list
> conda activate ambiente
> conda deactivate
```

Physalia Courses

NikolayOskolkov / **Physalia_EnvMetagenomics_2025**

Type / to search

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

Physalia_EnvMetagenomics_2025 Public

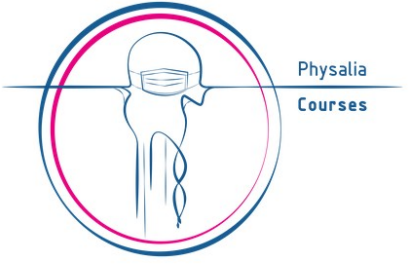
Pin Watch 0 Fork 1 Star 2

main 1 Branch 0 Tags Go to file t Add file Code

NikolayOskolkov Merge pull request #1 from AroneyS/main e6c71f0 · 1 hour ago 83 Commits

Articles	added Articles	11 months ago
Lectures	add binning, qc, annotation, catalogue slides	12 hours ago
LICENSE	added course material	11 months ago
README.md	Update README.md	3 days ago
command-line-basics.md	added course material	11 months ago
exercises.md	Merge pull request #1 from AroneyS/main	1 hour ago
physalia-logo.png	added course material	11 months ago
schedule.md	add binning, qc, annotation, catalogue slides	12 hours ago

README GPL-3.0 license



Environmental metagenomics

Instructors

- Dr. Nikolay Oskolkov, Lund University

About

No description, website, or topics provided.

Readme
GPL-3.0 license
Activity
2 stars
0 watching
1 fork

Releases

No releases published
Create a new release

Packages

No packages published
Publish your first package

Contributors 4

LeandroRitter Nikolay Oskolkov
NikolayOskolkov Nikolay Oskolkov
luispedro Luis Pedro Coelho
AroneyS Samuel Aroney

Note: All exercises will be executed inside the `Physalia_EnvMetagenomics_2025` folder that you cloned inside your own `home` folder.
So remember to `cd ~/Physalia_EnvMetagenomics_2025` every time you connect to the remote machine.

Getting the raw data

The data we are going to use originate from a public dataset and represent stool samples from modern infants from the DIABIMMUNE database (Three Country Cohort) from the Broad Institute.

You can download the raw data if you want as:

```
wget https://diabimmune.broadinstitute.org/diabimmune/data/16/G65860_pe_1.fastq.gz
wget https://diabimmune.broadinstitute.org/diabimmune/data/16/G65860_pe_2.fastq.gz
wget https://diabimmune.broadinstitute.org/diabimmune/data/16/G69146_pe_1.fastq.gz
wget https://diabimmune.broadinstitute.org/diabimmune/data/16/G69146_pe_2.fastq.gz
```

However in this course the data have been already downloaded for you and placed in the "Share" folder. Copy the raw sequencing data to your own `01_DATA` folder. Also copy the file `SAMPLES.txt`, which will be useful for running `for` loop on all the samples.

```
cd ~/Physalia_EnvMetagenomics_2025
mkdir 01_DATA

cp ~/Share/toy_data/*.fastq.gz 01_DATA/
cp ~/Share/toy_data/SAMPLES.txt ./
```

Let us now explore the data a little bit. First of all, we can look inside the gzipped-file without unzipping with `zcat`:

```
zcat 01_DATA/G69146_R1.fastq.gz | head
```

You should see 4 lines corresponding to each read: the first line contains the read ID (each starting with @), the second line corresponds to the sequence of the read, the third line is the delimiter and the fourth line contains ASCII quality scores for eac sequenced nucleotide.

Let us now count the number of reads in the fastq-files:

```
find 01_DATA -name '*R1.fastq.gz' | xargs zgrep -c ^@
```

How many reads do we have in the fastq-files?

QC and trimming

Now that you have copied the raw data to your working directory, let's do some quality control.
The sequencing process is subject to several types of problems that can introduce errors and artifacts in the sequences.
Because of this, bioinformatics analyses usually start with the quality control of raw sequences.
He we will use FastQC and MultiQC to obtain quality reports, and Cutadapt for trimming the Illumina data, respectively.